

Answer Key — Pressure in Liquids and Air

Final answer with a one-line reason. Full methods are in the Notes' worked examples.

Objective

- Q1** **(b) height of the liquid column** — liquid pressure depends on the height (depth), not the width.
-
- Q2** **(c) in all directions** — a liquid presses on the bottom and sides — in all directions.
-
- Q3** **(b) outside air pressure is higher than inside** — reduced pressure inside and higher atmospheric pressure outside hold it.
-
- Q4** **(b) to withstand the large water pressure near the bottom** — water pressure is greatest near the bottom, so a broad base resists it.
-
- Q5** **(b) False** — our internal pressure balances the atmospheric pressure.
-
- Q6** **atmosphere** — the air around Earth is the atmosphere.
-
- Q7** **millibar** — 1 millibar = 1 hPa = 100 Pa.
-

Short answer

- Q8** It acts in all directions (bottom and sides) and increases with the height/depth of the liquid column.
-
- Q9** The pressure exerted by the air of the atmosphere on all objects. We are not crushed because the pressure inside our body equals and balances it.
-
- Q10** The liquid exerts pressure on the side walls of the container, so water is pushed out through the side holes.
-

Long / application

- Q11** Liquid pressure depends only on the height of the column, not the width, so equal heights give equal pressure and equal bulges. Adding more water increases the height in that pipe, raising the pressure, so its balloon bulges more.
-
- Q12** Cover an inverted paper plate (with a stick) first with a folded chart sheet, then with a larger unfolded sheet, and lift each. The larger sheet is harder to lift though its weight is the same — air pushes down on it, and the force is greater over the larger area, showing air exerts pressure.
-